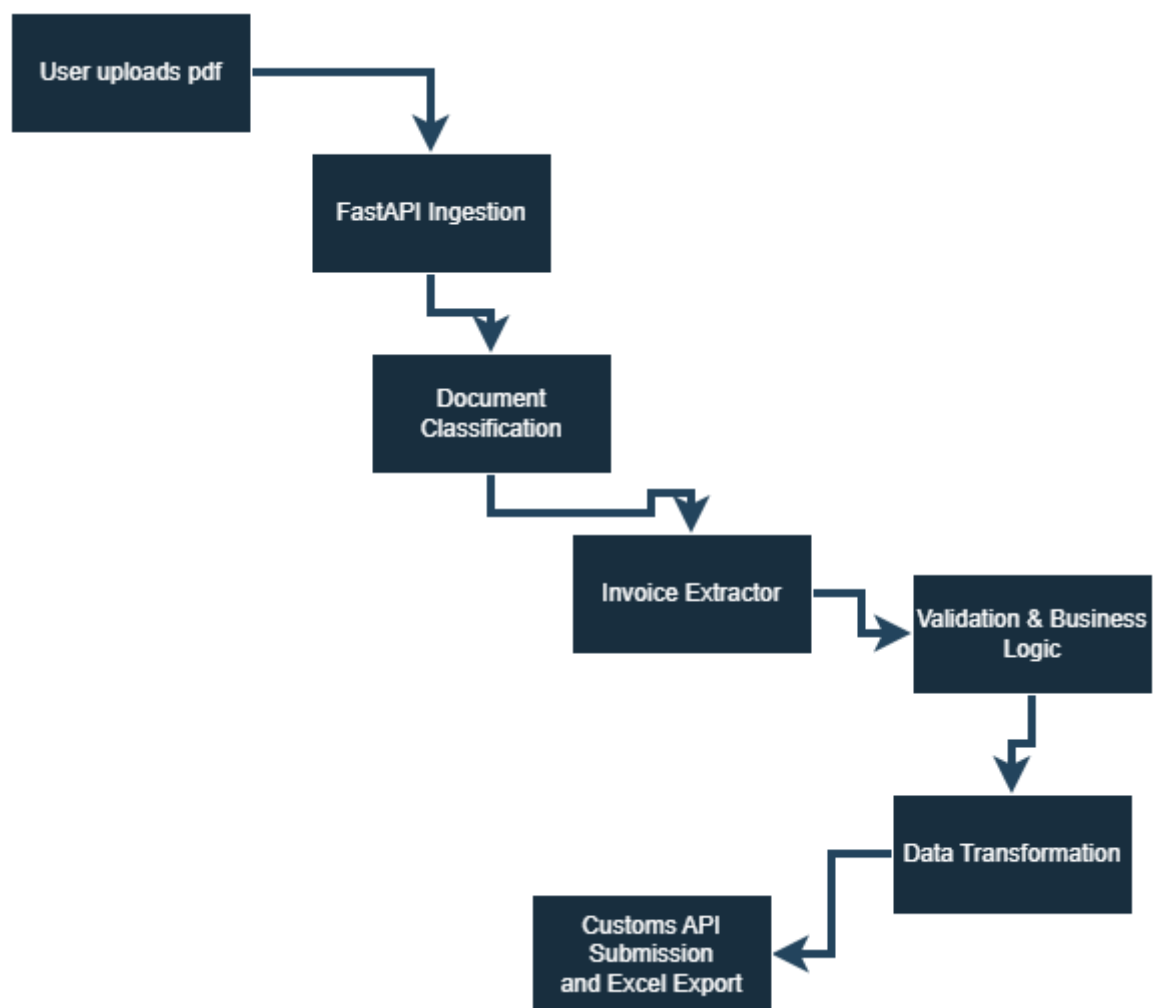


AI-Powered Logistics Document Processing System

AIM: The aim of this project is to design and build an AI-powered platform that automatically ingests logistics documents (such as Invoices and Bills of Lading), classifies them, extracts structured data, validates key business rules, transforms the data into customs-compliant formats, and prepares it for API submission to customs clearance systems and Excel export.

FLOW DIAGRAM:



Architecture Overview:

- Document Ingestion

The system ingests PDF logistics documents (Invoices, Bills of Lading, Packing Lists) via a FastAPI upload endpoint. Files are securely stored locally for processing. Both digitally generated and scanned PDFs are supported.

Technologies: FastAPI, Python, PDF files

- Document Classification

Uploaded documents are classified into Invoice, Bill of Lading, Packing List, or Unknown using a lightweight LLM.

Technologies: Ollama, LLaMA 3, pdfplumber

LLM Justification: LLaMA 3 provides accurate semantic classification while running locally, eliminating API costs and data privacy concerns.

- Data Extraction

Key fields and line items are extracted using rule-based parsing and table extraction. OCR is automatically triggered for scanned PDFs when text extraction fails.

Technologies: pdfplumber, Camelot, Tesseract OCR, Regex

Approach: Hybrid extraction (rules + tables + OCR fallback) for higher accuracy across document formats.

- Validation and Business Logic

Extracted data is validated for mandatory fields and structural correctness. HSN validation is currently bypassed due to the absence of a predefined reference dataset.

Technologies: Python

Logic: Mandatory field checks, line-item presence validation

- Data Transformation

Validated data is transformed into a structured, customs-compliant schema suitable for downstream systems and reporting.

Technologies: Pandas

Output Format: Structured JSON and tabular format

Future Enhancements & Improvements: Fine-tune a domain-specific LLM on logistics and customs documents to improve classification and field extraction accuracy. Introduce multi-language support for invoices and shipping documents. Build a dashboard for document status tracking, manual review, and corrections. Add role-based access control and document encryption. AI agent functionality can be added to make the platform autonomous by intelligently deciding extraction methods, validating data through reasoning, handling errors, and triggering API show only when human intervention is required.